Question	Answer	Marks	AO	Guidance	
7	**DR** $x^2 - 7x + 2 = 14 - 9x - x^2$	**M1**	**3.1a**	Attempt to eliminate y	
	$2x^2 + 2x - 12 = 0$ $(x + 3)(x - 2) = 0$	**B1**	**1.1**	Correct three term quadratic or factors seen	
	$x = -3, 2$	**A1**	**2.1**	Both correct values	
	Area $= \int_{-3}^{2}\left((14 - 9x - x^2) - (x^2 - 7x + 2)\right) \mathrm{d}x$ $= \int_{-3}^{2}(12 - 2x - 2x^2) \mathrm{d}x$	**M1***	**3.1a**	Attempts to integrate either function or the difference of two functions. Allow sign error and limits interchanged	Also allow for two separate integrals
	$= \left[12x - x^2 - \frac{2x^3}{3}\right]_{-3}^{2}$	**A1**	**1.1**	Correct indefinite integral	If done as two integrals, both need to be correct
	$= \left(12 \times 2 - 2^2 - \frac{2 \times 2^3}{3}\right) - \left(12(-3) - (-3)^2 - \frac{2(-3)^3}{3}\right)$	**M1(dep)**	**2.1**	Substitution of their limits into their expression seen and subtraction of results seen Allow for limits reversed or subtraction back to front Note: minimum evidence needed for full credit $\left(24 - 4 - \frac{16}{3}\right) - \left((-36) - 9 + 18\right)$	Must see substitution and subtraction in both integrals if done separately.
	$= \frac{125}{3}$	**A1**	**2.1**	**AG** From fully correct working Allow if $-\frac{125}{3}$ found if sign change justified eg "area is positive" Note $\int_{-3}^{2}(14 - 9x - x^2) \mathrm{d}x = \left[14x - \frac{9}{2}x^2 - \frac{1}{3}x^3\right]_{-3}^{2} = \frac{485}{6}$ and $\int_{-3}^{2}(x^2 - 7x + 2) \mathrm{d}x = \left[\frac{1}{3}x^3 - \frac{7}{2}x^2 + 2x\right]_{-3}^{2} = \frac{235}{6}$	
		[7]			